Assignment 2: Paper 3

β-VAE: Learning Basic Visual Concepts
with a Constrained Variational Framework

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1 Problem statement and understanding of the methodology

1.1 The problem and intuition

The authors ask whether a model can learn the factors that generate an image without being given their values [1]. In the 2D Shapes dataset, an object's appearance depends on its shape, horizontal and vertical position, scale and rotation. A useful representation would separate these factors so that changing a position coordinate moves the object while largely preserving its size and orientation. The paper investigates whether putting more pressure on the latent representation of a variational autoencoder helps produce this separation.

A VAE balances reconstruction against a penalty that keeps its inferred latent distribution close to a prior. The authors introduce a coefficient β on that penalty. Increasing β makes it more costly to encode information about each image. Their hypothesis is that a suitable restriction encourages the model to represent the independent factors more separately, although some image detail may be lost.

1.2 Mathematical formulation

Let $S = \{x_i\}_{i=1}^n$ contain images $x_i \in \{0,1\}^N$, with $N = 64 \times 64 = 4096$ pixels. Let $z \in \mathbb{R}^M$ be a latent vector, with $M = 10$ for this experiment. The encoder has parameters ϕ and produces an approximate posterior $q_\phi(z \mid x)$. The decoder has parameters θ and specifies the observation distribution $p_\theta(x \mid z)$. The prior is $p(z) = \mathcal{N}(0, I_M)$.

The lecture notes express the evidence lower bound as expected reconstruction log likelihood minus a KL penalty [2]. The paper changes the weight on that penalty:

$$\mathcal{L}_\beta(x; \theta, \phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \beta D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p(z)) \,.$$

Training maximises the average of $\mathcal{L}_\beta$ over images. Equivalently, it minimises its negative. The reconstruction term rewards retaining image information; the KL term penalises departures from the unit Gaussian prior.

The constrained interpretation is to maximise reconstruction quality subject to a permitted KL budget ϵ. For one image, the Lagrangian in the paper's equation (3) is

$$F = \mathbb{E}_q[\log p_\theta(x \mid z)] - \beta\big(D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)) - \epsilon\big), \qquad \beta \geq 0.$$

For fixed β and ϵ, dropping the constant $\beta\epsilon$ gives $\mathcal{L}_\beta$. This explains the penalty's role, but does not give a numerical value of β that guarantees disentanglement.

1.3 What changes when β changes?

Write $D(x) = D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)) \geq 0$. Holding the parameters fixed gives

$$\mathcal{L}_\beta(x) = \mathcal{L}_1(x) - (\beta - 1)D(x).$$

For $\beta > 1$, the objective decreases or stays unchanged and remains a lower bound on $\log p_\theta(x)$, since $\mathcal{L}_\beta \leq \mathcal{L}_1 \leq \log p_\theta(x)$. For $0 \leq \beta < 1$, the objective increases or stays unchanged, but it is no longer guaranteed to be a lower bound. At $\beta = 1$, it is the usual VAE ELBO. These comparisons hold at fixed parameters. Retraining changes the encoder and decoder, so reconstruction quality and disentanglement must be measured separately.

1.4 How the authors test it

Figure 7A/B compares $\beta = 4$ with $\beta = 1$ using the same fully connected architecture on 737,280 binary images. The images cover three shapes, six scales, forty rotations and 32×32 positions. Factor values are withheld from VAE training and used afterwards to inspect the representation.

The top rows show how inferred latent means respond to position, scale and rotation, averaging over the other factors. The lower rows show decoded images when one latent coordinate is varied and the remaining coordinates are fixed. In panel A, position and scale have selective coordinates, while two coordinates encode rotation. Panel B shows more mixing of factors. Section 4.2 describes the $\beta = 4$ representation as imperfect. The relevant comparison is the degree of factor selectivity; an exact one-coordinate-per-factor match is not required.

The paper also proposes a quantitative test: fix one factor in pairs of images, vary the others, and train a linear classifier to identify the fixed factor from average absolute differences in the inferred latent means. Figure 6 reports the resulting scores. Its β sweep shows that increasing the penalty indefinitely does not keep improving disentanglement. An excessively restricted representation can lose factors and produce poorer reconstructions. This reproduction covers Figure 7A/B only.

2 Reproduction of Figure 7A/B

The submitted figure uses the completed pair of models from the earlier IDE-assisted notebook. Both models trained for 100 epochs, and all results in this report come from that pair. The later ChatGPT conversation reviewed the methodology; it records ChatGPT Astra, with model identifier `gpt-6-astra-wm`.
Conversation: `https://chatgpt.com/share/6adadb56-35dc-83e9-9c94-7e1077a90301`
GitHub repository: `https://github.com/Jireh1226/AdvancedML_assignements`
Paper folder: `Assignment_2/paper_3_beta_vae/`
Submitted notebook: `beta_vae_learning_basic_visual_concepts.ipynb`
Submitted results: `results_beta_vae/seed42_v1/`.

2.1 Phase 1: understanding the paper and asking for code

The initial implementation request covered Figure 7A/B, restricted the work to the paper and lecture notes, and asked for failed attempts and corrections to be recorded separately from the finished notebook. Seed 42 was requested as a familiar starting point. It makes the experiment repeatable within the recorded setup, but there is no evidence that the authors used it or that it improves agreement with their figure.

The later chat asked the LLM to define the objective, explain its constrained interpretation and identify the architecture, activations and loss reductions for 2D Shapes. Answers had to point to the relevant equation, table or caption. This helped distinguish paper-specified settings from implementation choices. For example, the paper supplies Adagrad's learning rate, 0.01; deriving another rate from image dimensions or an assumed smoothness constant was unnecessary. The batch size, initialisation and training budget remained assumptions.

2.2 Phase 2: reviewing the implementation

The loss reductions were important to check. The reconstruction term sums over 4,096 pixels, the KL term sums over ten latent coordinates, and the combined objective is averaged over images. Averaging only the reconstruction over pixels would multiply the effective KL weight by 4,096. The submitted implementation uses the intended sums. The decoder's hidden layers use tanh, while its final layer returns linear logits for a Bernoulli likelihood. Applying tanh directly as a probability output would be invalid because it can return negative values.

Two presentation and portability problems were corrected without changing the learned weights. Fixed three-decimal variance labels hid small positive values as `0.000`; three significant figures preserve their scale. Cached summaries also contained absolute server paths, and an environment check blocked read-only inspection on another computer. Completed-result loading now resolves paths against the current result directory while retaining the original environment record. Incomplete runs still require a matching environment before resuming.

2.3 Phase 3: running, correcting, and comparing the results

2.3.1 Settings and objective

Both submitted models use all 737,280 binary dSprites images without augmentation or standardisation. Factor values are withheld from training and used only for analysis. There is no held-out split in this figure experiment. The data archive's checksum and the factor ordering were checked before training.

The encoder has two 1,200-unit ReLU layers followed by separate linear heads for ten means and ten log variances. The decoder has three 1,200-unit tanh layers and 4,096 output logits. This gives 14,196,516 trainable parameters. For a diagonal Gaussian posterior, a training sample is

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \xi, \qquad \xi \sim \mathcal{N}(0, I_{10}).$$

With decoder logit a_{ij} for pixel j of image i, the minimised batch objective is

$$J_\beta(B) = \frac{1}{|B|} \sum_{i \in B} \left[\sum_{j=1}^{4096} \mathrm{BCEWithLogits}(a_{ij}, x_{ij}) + \frac{\beta}{2} \sum_{k=1}^{10} (\mu_{ik}^2 + \sigma_{ik}^2 - 1 - \log \sigma_{ik}^2) \right].$$

One posterior sample per image estimates the expected reconstruction term during training. This follows the paper's Bernoulli observation model and the lecture's reconstruction–KL decomposition.

Specified in the paper	Implementation choice
Table 1 architecture, ten latents, ReLU/tanh and Bernoulli observations	Separate mean/log-variance heads; linear output logits; Xavier-uniform weights and zero biases
Adagrad, learning rate 0.01	Batch 256; initial accumulator 0.1; stabiliser 10^{-10}; no weight or learning-rate decay
Figure 7A/B: $\beta = 4$ and $\beta = 1$	Seed 42; matched initial weights and batch order; 100 epochs each; FP32 without TF32
Columns ordered by learned Gaussian variance; prior-range traversals	Dataset-mean posterior variance; all ten columns; one fixed oval; five values from +3 to −3

Table 1: Settings for the submitted pair. The paper does not establish the batch size, initial accumulator, seed, initialisation or epoch budget used here.

Each epoch contains $737280/256 = 2880$ updates, giving 288,000 updates per model. Initial-weight hashes agree across the two models. Both use the same per-epoch shuffle seed and reset the random state before training. The recorded epoch times sum to approximately 18.6 minutes for the pair, excluding data preparation and figure analysis. The framework versions and full configurations are saved with the results.

2.3.2 Failed attempts and checks

A four-example synthetic test produced extremely large KL values despite finite losses. A pilot using 500 real-data batches was more informative: with the original $\beta = 4$ objective, the initial spike subsided and the last pilot minibatch had reconstruction loss 233.79 and KL 7.50 nats per image.

Dividing the entire objective by 4,096 was then tested as a possible numerical adjustment. This preserves the nominal β and the objective's minimisers, but does not preserve Adagrad's trajectory with a nonzero initial accumulator. Its last pilot reconstruction loss was 2,828.94 rather than 233.79. The adjustment was rejected, and neither submitted full run uses it. A separate $\beta = 1$ pilot reached reconstruction 214.74 and KL 17.32 on its last minibatch. These pilot values are stability checks, not the reported final results.

Analytical checks verified reconstruction loss $4096 \log 2$ at zero logits, zero KL for a unit Gaussian posterior, and KL 5 for a ten-dimensional unit-variance posterior with every mean equal to one. Interrupted-versus-uninterrupted CPU tests checked continuation. After training, all images were re-encoded and factor-grouped averages were independently checked against the saved analysis. Recomputed traversal pixels differed by at most 1.67×10^{-6} for $\beta = 4$ and 2.29×10^{-6} for $\beta = 1$. Checkpoint hashes remained unchanged during these analysis and presentation checks.

2.3.3 How the figure is constructed

The position row averages each inferred mean over shape, scale and rotation at every position. The scale row averages over rotation and position, separately for each shape; the rotation row averages over scale and position. Colours follow the paper: square red, oval green and heart blue.

Columns are sorted by increasing dataset-mean posterior variance,

$$\overline{\sigma_k^2} = \frac{1}{n}\sum_{i=1}^{n} \sigma_{\phi,k}^2(x_i).$$

This differs from the variance of inferred means across images. The paper specifies ordering by learned Gaussian variance but does not state the aggregation rule. All ten coordinates are shown here; the original groups unused coordinates into one column.

The fixed reference image is an oval with zero-based factor indices $(0, 1, 3, 0, 16, 16)$ for colour, shape, scale, rotation and both positions. Its posterior mean supplies the fixed coordinates. Each traversal replaces one coordinate with $3, 1.5, 0, -1.5, -3$, in standard-normal prior units, and displays the decoded Bernoulli probabilities. The reference is the same for both submitted models and was not chosen separately to improve each panel.

2.3.4 Results and differences from the paper

Both models completed 100 epochs. Table 2 separates the online training measurements from reconstruction obtained by decoding the final posterior mean. The latter is deterministic and is not an estimate of the expected reconstruction term in the ELBO.

Measurement, nats per image	$\beta = 4$	$\beta = 1$
Last-epoch sampled reconstruction	54.076	36.091
Last-epoch KL	11.889	17.603
Final posterior-mean reconstruction	45.145	33.456
Final dataset-mean KL	11.900	17.606

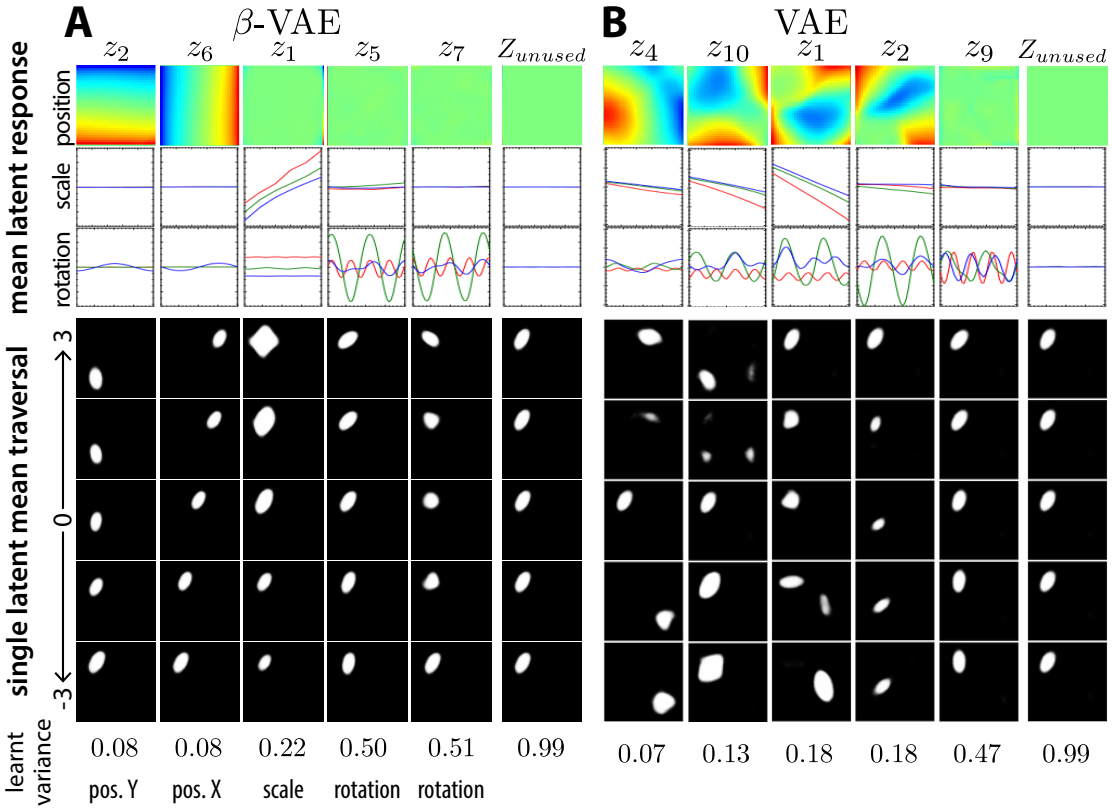
Table 2: Measurements from the completed seed-42 pair. Last-epoch values average batches while parameters change; final values use the fixed saved models.

The stronger KL penalty gives a lower KL and higher reconstruction error. This is consistent with restricting the information in the representation. The response maps also show a qualitative difference: for $\beta = 4$, coordinates 10 and 3 mainly represent the two positions, coordinate 7 responds to scale, and coordinates 4 and 9 respond to rotation. In the $\beta = 1$ model, the position responses are more mixed across coordinates 2, 7 and 6, and traversals change several properties together. Both models have five low-variance coordinates and five approximately prior-like coordinates. That count alone cannot distinguish disentanglement from entanglement.

Figure 1 supports the paper's qualitative contrast, with visible differences. The $\beta = 4$ position traversals are relatively selective, but the scale and rotation columns also change shape in some rows. Thus the reproduced representation remains imperfect, as the paper acknowledges for its own example. The five informative posterior variances are approximately 0.0028, 0.0029, 0.0196, 0.0767 and 0.0940, below the original panel's displayed 0.08, 0.08, 0.22, 0.50 and 0.51. These numbers do not establish a better representation: a smaller posterior variance says how uncertain the code is, not whether it isolates a semantic factor.

The paper does not identify the exact seed, initialisation, batch size or training duration behind these panels. Those choices, the unspecified Adagrad accumulator and the reference image can affect

Original Figure 7A/B



Completed seed-42 reproduction

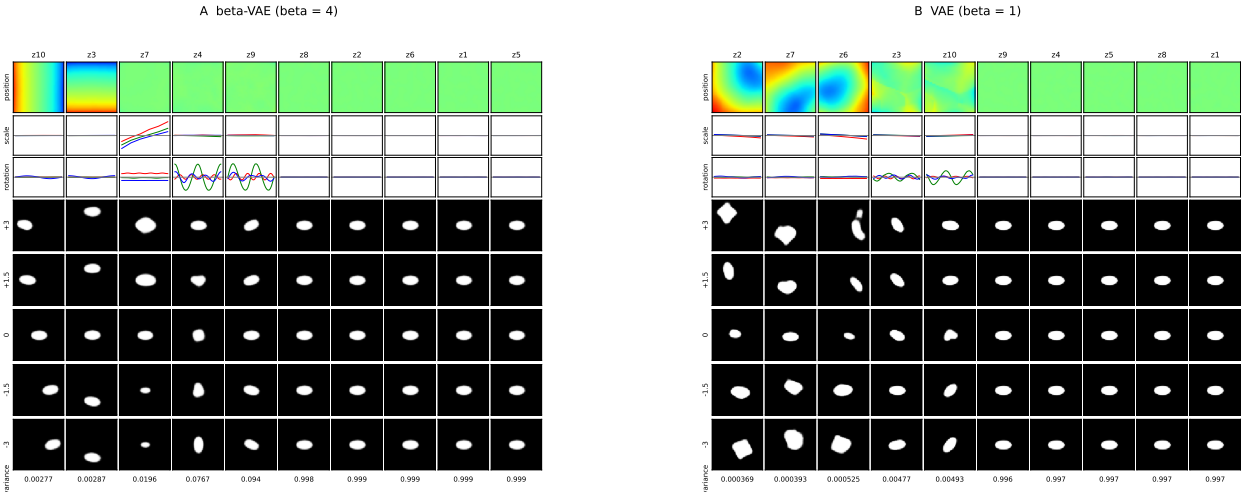


Figure 1: Original panels from Higgins et al. [1] above and the submitted reproduction below. Both compare $\beta = 4$ (A) with $\beta = 1$ (B). Coordinate identities can change between runs; the comparison concerns factor selectivity in the response rows and traversals.

the learned coordinates and their decoded appearance. Their individual effects were not isolated here. Coordinate numbers and signs can change without altering factor selectivity, so matching their identities was not a target.

The fixed budget also limits the comparison. From epoch 80 to 100, sampled reconstruction fell from 59.54 to 54.08 for $\beta = 4$ and from 38.74 to 36.09 for $\beta = 1$. Training was still improving; 100 epochs is not evidence of convergence. One matched seed illustrates the reconstruction–disentanglement trade-off but cannot establish how reliably it occurs. No Figure 6 classifier score or multi-seed robustness claim is inferred from these plots.

2.4 Phase 4: comparison with a released implementation

The completed earlier notebook was saved in local Phase 3 commit `8b28d5b` before this source check. The authors' publication page links the paper and the official dSprites dataset [3]. The linked dataset repository provides the images, metadata and a reloading example, but no Figure 7 VAE training script was identified in the checked resources. No third-party training implementation was opened or substituted for an author release.

The dataset documentation confirms the complete factorial grid used here. It does not resolve the missing optimiser accumulator, batch size, initialisation or stopping rule. A line-by-line training-code comparison therefore could not be completed. This is a limitation of the located material, rather than evidence that no author implementation exists elsewhere. The checked URLs and the saved notebook identity are recorded in `phase4_record.json`.

2.5 Code and interaction record

The submitted notebook and its completed models preceded the shared chat and were produced with Codex/IDE assistance. The chat subsequently checked the paper and proposed another implementation. It does not preserve the interaction that produced the submitted figure, so this workflow does not fully meet the assignment's chat-interface requirement. The completed IDE-generated result is identified directly rather than attributed to the later chat.

The notebook retains its executed outputs; configurations, histories, analysis arrays and figures accompany it in the repository. Large checkpoints and the dataset are excluded. Run instructions explain data preparation and fresh training. The original dataset access was limited to the archive, while the publication page and dataset listing were checked later for Phase 4. The saved change log records failed pilots and corrections; a separate finalisation record identifies the completed run used throughout this report.

3 Critical review

3.1 Assessment of the paper

The paper's strongest contribution is a simple experiment that connects a change in the VAE objective to an interpretable change in the representation. The $\beta = 1$ and $\beta = 4$ models use the same 2D Shapes architecture, making the role of the penalty easier to examine. Figure 7 combines factor-response plots with decoded traversals. The response plots show what the encoder records, while the traversals show how the decoder uses each coordinate. This is more informative than judging disentanglement from reconstruction quality alone.

The constrained formulation gives a useful intuition, but the link from prior matching to semantic factors remains incomplete. The lecture's non-negativity of KL explains why matching a prior is a penalty. It does not show that a latent coordinate must correspond to scale or position. A diagonal Gaussian posterior already assumes conditional independence between coordinates for a fixed image, even at $\beta = 1$. Across images, the inferred means can still vary together. The paper itself distinguishes statistical independence from alignment with interpretable factors in Section 3, and acknowledges in Section 4.2 that alignment with human-intuitive axes is not guaranteed. The empirical comparisons support the method in the tested settings; the Lagrangian argument alone does not establish that stronger prior matching will recover the intended factors.

The quantitative evaluation is useful but its reporting weakens the robustness claim. Appendix A.4 trains ten model replicas and evaluates each with three classifier seeds, giving thirty scores. It then discards the lowest fifteen scores to remove outliers, including divergent runs. Removing half the scores changes the question from typical performance across attempts to performance among the selected higher-scoring evaluations. These thirty scores also come from only ten encoder initialisations; reseeding the classifier does not produce a new encoder. Reporting all scores, the number of failed encoder runs and variation between encoders would make the stability claim easier to assess. The paper's reported 99.23% score for β-VAE and 61.58% for VAE in Figure 6 should be read with this selection rule in mind.

The metric's scope also matters. Its classifier identifies which of four factors was fixed: horizontal position, vertical position, scale or rotation. Shape identity is a fifth factor in the dataset but is not a target of this particular metric. Appendix A.5 separately tests prediction of all five factor values, which adds evidence about the information retained in the representation. As the authors note, predicting factor values does not by itself measure their independence. A high score on one of these tests cannot settle every part of the disentanglement claim.

Figure 7 has related limits. Averaging across other factors can hide interactions, and traversing one reference image only shows how the decoder behaves around that image's inferred code. The same coordinate could behave differently for a heart or square. Rotation needs particular care because the rendered square repeats every $\pi/2$, the oval every π, and the heart every 2π. The paper observes two rotation-related coordinates with responses resembling sine and cosine. Two such coordinates need not indicate failure: the angular factor is periodic, and different shapes have different visible symmetries. Further evidence could come from several prespecified reference images and response curves conditioned on the other factors, without choosing examples after seeing their appearance.

The clean factorial dataset makes this analysis possible, but also limits what it establishes about generalisation. Every combination of the sampled factors is present, and the assigned figure does not test new combinations or correlated sampling. The paper also includes qualitative experiments on faces and chairs, so its evidence extends beyond synthetic shapes. Those examples still leave open how consistently factor separation survives when correlations between attributes change. A

controlled test could hold out selected factor combinations and measure whether varying a learned coordinate preserves the other attributes on those held-out combinations.

The reconstruction trade-off is another reason to avoid interpreting a larger β as automatically better. A representation can become simpler by losing useful information. If the posterior becomes almost identical to the prior for every image, the decoder receives little image-specific information, even though the KL is small. The paper's β sweep and blurry reconstructions acknowledge this cost. Comparing selective traversals alongside reconstruction and per-coordinate KL would help distinguish useful factor separation from discarded factors. One seed in the current reproduction can illustrate either behaviour, but cannot establish which is typical.

The paper supports β-VAE as a practical way to encourage disentanglement, with a measurable cost in retained image information. Its strongest evidence is the controlled VAE comparison. The main reservations concern the selection of reported scores, the limited coverage of the metric and the absence of a general guarantee that the learned coordinates align with the intended factors.

3.2 Comparison with the OpenReview discussion

Reviewer 1 also valued the method's simplicity and questioned discarding the bottom half of the scores. The authors attributed this exclusion to divergent runs. That explanation leaves the selection concern above unresolved. Reviewer 3 questioned sampling density, the limited traversal examples and whether reference images were held out. The authors confirmed that those images came from the training data [4].

Reviewer 2 placed more weight on likelihood estimates and generated-image quality. This assessment instead asks whether the representation separates useful factors, while treating reconstruction as a cost that should be measured. Reviewer 2's suggested comparison with a smaller latent-space VAE is a useful additional control; the authors reported that reducing latent size lowered their disentanglement score [4].

The four-factor metric's coverage and the shared encoders behind repeated classifier evaluations add more specific qualifications to these concerns. The comparison uses archived original OpenReview records because the live forum required browser verification. Official reviews are distinguished from public comments and author responses.

References

[1] Irina Higgins, Loic Matthey, Arka Pal, Christopher Burgess, Xavier Glorot, Matthew Botvinick, Shakir Mohamed and Alexander Lerchner. β-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework. ICLR, 2017. `https://openreview.net/forum?id=Sy2fzU9gl`.

[2] Hirak Sarkar. CS 6362 Advanced Machine Learning, Fall 2026. Lecture notes, 23 September 2026, `hs_wednesday_Sep_23.pdf`, pp. 22–26.

[3] Loic Matthey. Author publication page for β-VAE. `https://matthey.me/publication/beta_vae/`. Google DeepMind. dSprites: Disentanglement testing Sprites dataset. `https://github.com/google-deepmind/dsprites-dataset`. Resources checked 22 October 2026.

[4] OpenReview. Reviews and author responses for paper `Sy2fzU9gl`. Official reviews: Reviewer 1, `HyRZoSLVe`; Reviewer 2, `H16z7IT4l`; Reviewer 3, `B1-0khZEl`. Author responses: `By7JdqZLx`, `HJrTOqWUx` and `B1vdd9-8g`. Archived original records: `https://github.com/xcfcode/openreview_spider/blob/b232dbcbbee6a7a39415c2ea046058e9dcd82e1e/res/Sy2fzU9gl.json`.